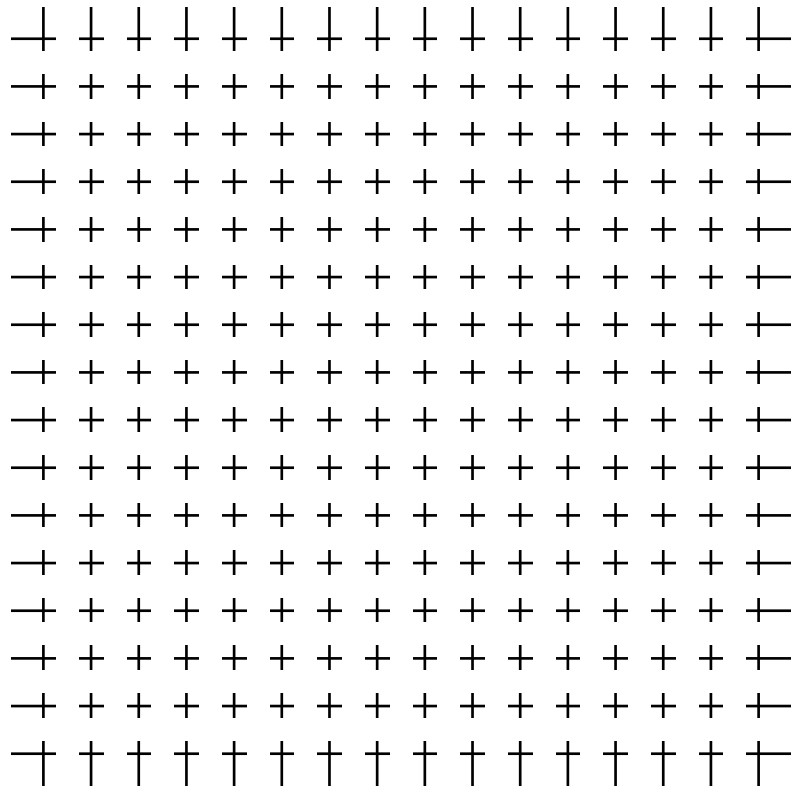




PLAYING WITH SYSTEMS

REFLECTIONS ON DESIGNING MEANINGFUL
LEARNING GAME EXPERIENCES

PLAYING WITH SYSTEMS –
REFLECTIONS ON DESIGNING MEANINGFUL
LEARNING EXPERIENCES

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ABSTRACT

Global problems are largely systemic in nature. Fixing systems is often close to impossible, because they are resilient to change and actual solutions are often counterintuitive and hard to pinpoint due to their complexity. The world needs people with the understanding and tools to effectively redesign these systems.

First part of this thesis presents the concept of systems learning games, whose purpose is to train players to understand and operate within complex systems. Systems learning games are games modeled to represent real-world systems, enabling players to experience systemic problems and the act of solving them first-hand in a controlled and abstracted setting. Optimally they focus not only on passing on knowledge, but also on supporting the development of right kinds of skills, character and meta-learning required to solve systemic problems in the real world.

While such systems learning games already exist, they largely ignore player-created meaning, that is often at the core of successful game experiences. The second part reflects on systems learning game prototypes I designed to try and implement gameplay that is simultaneously meaningful to the players and that allows players to achieve set learning goals. These reflections revolve around two central themes: arguments and complexity.

To achieve meaningful gameplay, learning goals can be converted into arguments. They are statements that are meaningful in the context of playing the game and the game world. These statements can be mapped to actual game mechanics using the Mechanics-Dynamics-Aesthetics framework. (Hunicke, 2004) The implementation of the arguments can also happen through worldbuilding, goals, theme or tone of the game.

In the context of modeling real-world systems to games, meaningful gameplay can also be achieved by honing in the right level of complexity. Game systems should be sufficiently complex to generate meaningful decision-making but simple enough in terms of the amount of detail to be maluable and understandable by the players.

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ABSTRAKTI

Maaailmanlaajuiset ongelmat ovat pääosin systeemisiä. Systeemien korjaaminen on usein lähes mahdotonta, koska ne ovat vastustuskykyisiä muutoksille ja todelliset ratkaisut ovat usein epäintuitiivisiä ja vaikeasti huomattavissa systeemien kompleksisuuden takia. Maailma tarvitsee ihmisiä, joilla on ymmärrys ja työkalut systeemien tehokkaan uudelleensuunnitteluun.

Työn ensimmäinen osa esittelee systeemioppimispelien konseptin. Niiden tarkoituksena on kouluttaa pelaajia ymmärtämään ja toimimaan kompleksien systeemien kanssa. Systeemioppimispelit ovat pelejä, jotka ovat mallinnettu oikean maailman systeemien pohjalta, antaen näin pelaajille mahdollisuuden kokea ja ratkaista systeemisiä ongelmia kontrolloidussa, abstraktoidussa ympäristössä. Optimaalisesti ne eivät keskity pelkästään tiedonvälitykseen, vaan auttavat pelaajia kehittämään oikeanlaisia taitoja, luonteenpiirteitä ja oppimaan oppimisen taitoja, joita tarvitaan oikean maailman systeemiongelmien ratkaisemisessa.

Vaikkakin tämänkaltaisia systeemioppimispelejä jo on, ne jättävät pääosin huomiotta onnistuneen pelikokemuksen ytimessä olevien pelaajan itsensä luomien merkitysten tärkeyden ja sen sijaan keskittyvät ainoastaan oppimistavoitteiden saavuttamiseen. Toinen osio pohtii pelinsuunnitteluprosessia kehittämieni systeemioppimispeliprototyyppien kautta, joissa yritin saavuttaa pelikokemuksia, jotka ovat merkityksellisiä pelaajalle ja samaan aikaan saavuttavat oppimistavoitteet. Näiden pohdintojen pääteema ovat väitteet ja kompleksisuus osana pelisuunnittelijan työkaluarsenaalia.

Merkityksellisen pelikokemuksen saavuttamiseksi oppimistavoitteet voidaan kääntää väitteiksi. Ne ovat toteamuksia, jotka ovat merkityksellisiä pelimaailman ja pelin pelaamisen kontekstissa. Nämä väitteet voidaan kartoittaa vastaaviin pelimekaniikkoihin käyttäen hyödyksi ”Mechanics-Dynamics-Aesthetics” työkalua. (Hunicke, 2004) Väitteiden tuominen peliin voidaan toteuttaa myös pelimaailman rakentamisen, pelitavoitteiden, teeman tai sävyn kautta.

Mallinnettaessa oikean maailman systeemejä peleihin, merkityksellisen pelikokemuksen voi saavuttaa myös hienosäätämällä pelin kompleksisuuden taso kohdalleen. Pelisysteemin tulisi olla tarpeeksi kompleksi, jotta se tuottaa merkityksellisiä päätöksentekohetkiä, mutta samaan aikaa tarpeeksi yksinkertaistettu yksityiskohtien määrän suhteen, jotta peli pysyy muovautuvana ja ymmärrettävänä pelaajille.

TEKIJÄ	EETU LEPPÄLÄ	TYÖN NIMI	PLAYING WITH SYSTEMS – REFLECTIONS ON DESIGNING MEANINGFUL LEARNING GAME EXPERIENCES
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1 INTRODUCTION

As we progress through the 21st century, it is becoming increasingly clear that most of our biggest problems on this planet are systemic. Hunger, poverty and war, as well as tribalism on the societal side, climate change, and biodiversity loss on the environmental side. These facts are bombarded to us via the news media every day, and to many it creates an overwhelming feeling of dread and apathy rather than a sense of being informed. It often leaves the person with a feeling that nothing can be done to solve these global calamities. (McNaughton-Cassill, 2001)

Understanding systems is necessary to be able to correct these systemic problems. I'm convinced that games are one robust way of educating the public, both children and adults, on the risks posed by inadequate systems design, but also on the act of systemic change and the capabilities necessary for such changes.

This train of thought led me to create game prototypes exploring the topic of systems learning in games in the Spring of 2022. As the prototypes took shape, the core ideas for this thesis also started to formulate.

CONTEXT

This thesis was driven firstly by my personal experiences with games. I've found that games have taught me many capabilities that help me cope with complexity, uncertainty and decision-making under pressure that have proven useful in the real world. I've long been curious about researching this topic deeper. Secondly, there is a growing body of work utilising games for learning, but it seems they are ineffectual. Thirdly, I feel that as the world changes, the methods with which we teach our children must also adapt. There is significant evidence that games can help provide the necessary skills for the coming century. (Castronova & Knowles, 2015; Dieleman & Huisinigh, 2006; Fabricatore & Lopez, 2012; Gatti Junior et al., 2020; Gee, 2009; Goodwin & Franklin, 1994; Kaneda et al., 2021; Katsaliaki & Mustafee, 2012; Mazur-Stommen & Farley, 2016; Ponte et al., 2015; Pope, 2021; Schrier, 2018; Stanitsas et al., 2019; Tsai et al., 2021) And lastly, the world has a wide variety of systemic problems, and we need people with tools to understand and design better systems that holistically take into account the peculiar ways systems operate.

OVERVIEW

This thesis is structured in two main parts. Chapter 2 is an extensive literature review where I dissect the concepts of systems, learning and games and their combined power to change behaviour and encourage action. In Chapter 4, I reflect on my experiments and observations from the perspective of connecting the three topics in a way that induces play and meaning in the players and thus allows learning within games.

RESEARCH QUESTIONS AND AIMS

In their book "Game Design Research", Lankoski & Holopainen (2017) outline various categorisations of game design research based on how they relate to the studied domain of knowledge and the applied methodology. (Lankoski & Holopainen, 2017) Regarding knowledge domains, they borrowed from Nigel Cross's definition of design research. (Cross, 1999) They have defined three categories of design research: design epistemology, design praxiology and design phenomenology. Within this framework, this work is mainly concerned with game design praxiology: how specific outcomes can be achieved systematically instead of with the actual game artefacts themselves.

Another way Lankoski & Holopainen propose dissecting design research is by examining its goals and approaches. Forlizzi et al. (2009) suggest three outcome-oriented categories: research on design, research for design and research through design. (Forlizzi et al., 2009) In this thesis, I research games through design. The prototypes I've built are the primary source material for the insights in Chapter 4.

As with a lot of design research, research questions and aims change according to the new findings discovered during design practice. As I made more prototypes and researched the topic of educational and learning games, I noticed that while there is literature on how games educate players, few research projects explore the specifics of how this is done in a meaningful way for the players. For this thesis, I wanted to focus on the following:

- How to design arguments that players will find meaningful, be they learning goals or other arguments about the world?
- How to design play spaces that encourage players to build capabilities?
- How to design games that generate understanding and skill to thrive in complex systems?

2 SYSTEMS, LEARNING AND GAMES

I have built this thesis on the concept of systems learning games.
To define it, I individually explore systems, learning and games
and highlight their connections.

2.1 SYSTEMS

2.1.1 NATURE OF COMPLEX SYSTEMS

Donella Meadows (2009) explains in her seminal book “Thinking in Systems: a primer” that systems are a collection of things interconnected in a way that produces their own behaviour. They can be dissected into elements, interconnections and functions. (Meadows, 2009, 2) A game of football is a system. The teams within the game are subsystems of the game. The players, the referee, the ball, the audience and the playing field are some elements within this system. Players in a team are interconnected. The two teams are also interconnected with each other. Each player has a function within their team. The referee’s function is to administer that rules are being followed. The function of the ball is to be kicked.

The units of a system interact to serve their function and, as an outcome, “exhibit adaptive, dynamic, goal-seeking, self-preserving and sometimes evolutionary behaviour” that is not intrinsic in any of the system parts. They are more than the sum of their parts. (Meadows, 2009) Players within the football team explore the space of possibilities to score goals with the ball. Tactics change according to opponent manoeuvres. As the team plays more games, it will gradually evolve into a better-functioning unit. None of these behaviours are described in the rules of football.

While systems can be dissected and separated from their surroundings for analysis, there are no actual boundaries between systems in the real world. Everything is interconnected. (Meadows, 2009) Similarly, in football, the game is interconnected with other systems. Games are played for large sums of money with big sponsors who themselves are outside of sports, hoping to gain brand awareness through associating with this particular team; games are televised, around which people have created other cultural phenomena; and football is tied into politics, and games between rival nations can be seen as political flexing.

In many cultures, we like to put things in boxes and separate phenomena from each other, including systems. In reality, however, they are continuous. This misunderstanding of the fundamental nature of systems creates a lot of friction, for example, at national borders. (Meadows,

2009) People within one cultural framework living on both sides of the border can create tensions relating to who actually “owns” the land.

This lack of boundaries, combined with the fact that systems are dynamic, ever-changing collections of things where interconnections are separated and formulated constantly, makes real-world systems complex. Complexity is a delicate balance between order and chaos (Adams & Dormans, 2012) and it creates a lot of benefits for systems: it makes them more resilient to unforeseen events and allows them to self-organise, creating stability in the long-term. However, Jay Forrester, the father of systems analysis, has described complex systems also as counterintuitive. (Meadows, 2009) Complexity, and the human inability to understand it, are significant sources of systemic problems.

2.1.2 SYSTEMIC PROBLEMS AND INTERVENTIONS

Hunger, poverty, war and other global maladies can be characterised as systemic problems. They are a sum of innumerable factors that may individually seem harmless. Meadows argues they are not deliberately created, nor does anyone want these problems to persist. Instead, they emerge from the system’s properties and the way the system is put together. (Meadows, 2009, 4) They are not a byproduct; the system is set up in a way that causes these issues.

Because most real-world systems are complex, problems may be unforeseeable before the system, and the expensive infrastructure to actualise it has been set up. Systems are not simply built overnight; they accumulate over decades on top of earlier iterations. Meadows states that systemic problems do not give in until the system they are part of is restructured. (Meadows, 2009, 4)

Instead of restructuring systems, systemic problems are often attempted to be solved with interventions. But often, due to the counterintuitiveness of complex systems, interventions direct the system behaviour in the opposite direction than what was intended. (Meadows, 2009)

Meadows outlines an example of a burden-shifting scheme. In it, the intervenor takes on some of the burden experienced by the system, monetarily or otherwise. While this scheme often alleviates the burden experienced by the system in the short term, it can also hide the actual state of the system and the root cause behind the prob-

lems. Instead, this same solution will need to be applied again and again. This may eventually lead to a situation where the system cannot survive and self-correct without help and becomes dependent on the intervenor. Ultimately, the intervenor may be unable to help in the needed capacity, and the system becomes significantly less effective or collapses. She gives an example of pest control in monoculture farming. There may be an urge to apply pesticides instead of inspecting if the monoculture farming methods have contributed to the imbalance of the local ecosystem and the outbreak of the pest problem. This will create a reinforcing feedback loop where the root problem deepens, and more pests require more pesticides, creating additional problems. (Meadows, 2009, 133)

Solving systemic problems is not the same as solving everyday problems. They require a particular understanding of the world and knowledge of systemically high-leverage methods to enact change. Instead of attempting to fix systems, which Meadows states is impossible because of the ability of complex systems to correct themselves into their original state, they need to be redesigned from the ground up. (Meadows, 2009) The very fundamentals of systems need to transform to enact lasting change.

Meadows has created a list of systems leverage points created for designing complex systems. These are illustrated in Figure 1. High leverage actions include creating rules, structures and goals of the system. While they have the potential to impact things drastically, they also require immense amounts of effort. The bigger the system, the more there are stakeholders that need convincing. Meadows also points out that while the systems leverage list is a work of years of rigorous study and experience with complex systems, they are not infallible. Similarly to systems themselves, leverage points are also counterintuitive. Complex systems cannot be, in her mind, generalised. Instead, these tools can be understood as a partial vocabulary for systems knowledge and implemented with a deep understanding of the specific context in question. (Meadows, 2009)

2.1.3 SYSTEMS ANALYSIS

Meadows argues that systems thinking is deeply imbued in traditional wisdom and exemplified by sayings such as “The rich get richer”

or “Don’t put all your eggs in one basket”. (Meadows, 2009, 4) This demonstrates that humans have built an intuitive understanding of systems and their problems.

She further claims that although we as a society have benefitted from the advances of logic and science, their success has favoured the attitudes of reductionism over holism and therefore made us instinctively look for single external solutions in the form of a pill or technical fix to systemic problems. This has inevitably prolonged a shift in attitudes toward holistic thinking required for systemic understanding. (Meadows, 2009)

Researchers with a modern understanding of systems as interconnected wholes have created ways of simulating them in closed environments, either as a computational model or a mathematical formula. These tools have been used to simulate systems to a high degree of accuracy, although the fact remains that a model is necessarily a simplification of a complex real-world system. The world contains an infinite number

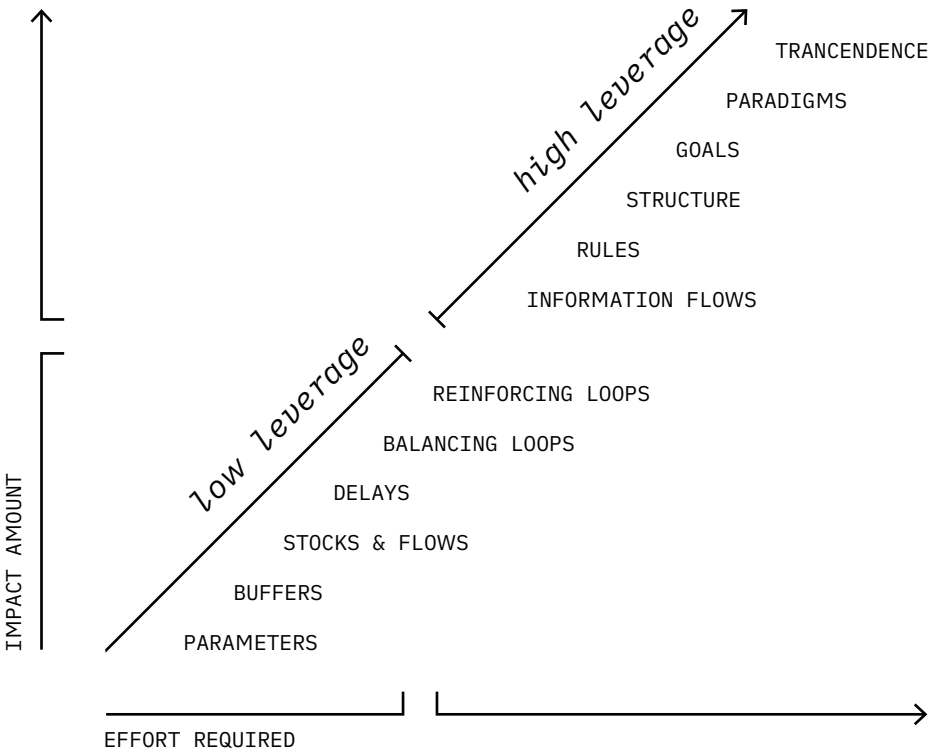


Figure 1. List of systems leverage points, impact amount of a leverage as a function of effort required (Meadows, 2009)

of variables that may have slight effects on the way things play out. There is no computational possibility to simulate complex systems to the degree that they would be 100% accurate. Thus, the model’s creator must decide what variables are important enough to be included and what degree of accuracy is acceptable for the system to remain valid. This is illustrated in Figure 2. Will Wright (2003) uses the metaphor of quantum physics and the general theory of relativity. For physicists, they are indispensable tools to describe and analyse physical phenomena, but as mathematical models, they are incompatible with each other; neither model can validate the other. (Lessons in Game Design, Lecture by Will Wright, 2003)

Various schools of thinking focus on the holistic understanding of systems, such as systems thinking (Meadows, 2009; Senge, 1990), systems intelligence (Saarinen & Hämmäläinen, 2004) or wicked problems (Rittel & Webber, 1973). Rigorous mathematical models have been developed over the years to try and model the exact behaviour of systems to predict their patterns of progression, such as cellular automata, systems dynamics, chaos theory, complex adaptive systems and graph theory, with contributions from people such as of Stanislaw Ulam, John von Neumann and Jay Forrester. Outside of science, there have been attempts trying to utilise systemic thinking, such as the “mental models” movement, with works such as General Thinking Concepts by Parrish (2019), Super Thinking by Weinberg (2019) and others mostly originating from talks given by Charlie Munger, a co-founder of Berkshire Hathaway. (Parrish et al., 2019; Weinberg & McCann, 2019)

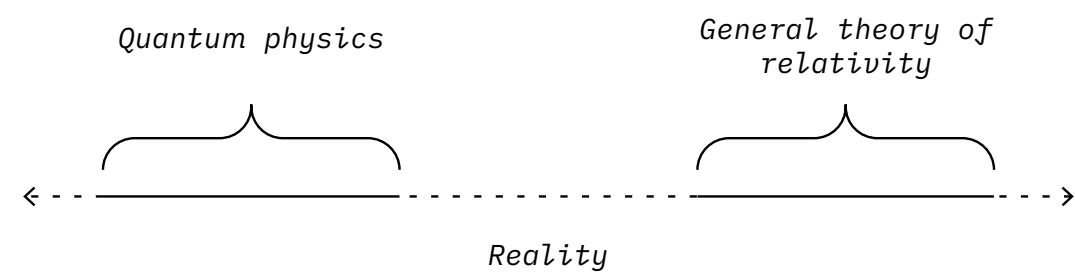


Figure 2. Incompatibility of models. (Lessons in Game Design, Lecture by Will Wright, 2003)

Each describes a point of view on the conundrum of systems and systemic problems. They are less like doctrines that need to be followed and more like tools, and a proficient systems analyst can use the right tool for the right job.

2.1.4 WIDESPREAD SYSTEMS UNDERSTANDING

Systemic thinking is a tool that could benefit everyone. It can help reveal deep patterns across complex systems and prevent reactionary behaviour. It can promote deeper analysis and empathy towards all people, including those working towards benefitting established harmful systems. This is exemplified by how many people studying complex systems talk about them in deeply empathetic and emotional terms. (Meadows, 2009; Peter Senge, 2014; Hunter, 2013; Dieleman & Huisinigh, 2006; Saarinen & Hämmäläinen, 2004) They emphasise the unity of all people across national borders and the value of all life, human or otherwise. By developing an intuition toward thinking systemically, there is potential to grow more nuanced, analytical and empathetic individuals.

Unfortunately, this topic remains convoluted and misunderstood. While there are signs that the general public is waking up to the dangers of systemic problems, the level of understanding remains largely on the level of traditional wisdom. Individuals and companies still get blamed for things they do instead of us being able to see past that to the underlying systemic errors allowing such behaviour.

There is a dire need for individuals who understand the core logic of these issues. Yet systems theory or systems thinking offers no solutions on how to activate passion in people towards these issues or how to train appropriate responses to system failures and enact them. Dieleman and Huisinigh (2006) note that not enough is being done to teach action towards systemic changes. Instead, systems topics are only taught from the perspective of understanding systemic issues. (Dieleman & Huisinigh, 2006) Wu and Lee (2015) suggest that first-hand experience dealing with issues sparks emotional triggers necessary for behavioural change. (Wu & Lee, 2015) This could be achieved by utilising experiential learning.

2.2 SYSTEMS LEARNING

2.2.1 CAPABILITIES

Systems learning could be categorised under a broader umbrella of “21st-century capabilities”. Fadel et al. (2015) outline a learning model that supports learning these capabilities. They state that current educational models are insufficient to support students in learning essential things to thrive or survive in the coming decades. They say that instead of only focusing on knowledge acquisition, there should be more focus on all types of capabilities: knowledge, skills, character and meta-learning. (Fadel & Groff, 2019) These are illustrated in Figure 3.

All these types of capabilities are also crucial in the understanding and redesign of systems. Knowledge of the history, dynamics and context of the systems observed is essential. The cultural and historical context and current agents within the system define it and the ways to influence it. Skills for making change within systems, such as utilising and improving intervention methods, facilitating negotiations and organising work, are important. Character traits that are well-suitable to working with systems are also learnable. Systems change is not something that may even happen during the lifetime of the practitioner. Mental toughness is required as also empathy and other character traits. Systems also have their own quirks that require a new type of meta-learning. Complex systems adapt and thus something that worked yesterday, might not work today, requiring us to view learning from phenomena and experiments differently.

2.2.2 SIGNATURE PEDAGOGIES

Lucas and Spencer (2013) outline a concept of signature pedagogies that can be used when deciding on the optimum type of teaching method for any learning content, visualised as a set of axes in Figure 4. The same style of teaching is not ideal for all types of learning. (Lucas & Spencer, 2018)

As mentioned earlier, Wu and Lee suggest that the signature pedagogies that favor experiential learning could be used effectively in the

context of learning about complex systems. (Wu & Lee, 2015) Because complex systems are unknowable and dynamic, the best way to get to know them is by experimenting with them. Dieleman and Huisinhg also argue that systems change is inherently about action and that action should be trained for. (Dieleman & Huisinhg, 2006)

Knowledge of complex systems can only be accrued by interacting with them. Skill manipulating them comes from training with them. Characteristics and meta-learning skills also develop through repeated exposure to complex systems and their peculiarities. This suggests that experiential learning could be optimal in the case of systems learning for all four types of capabilities. Experiential learning also allows for the development of all types of capabilities simultaneously compared to a traditional classroom learning situation.

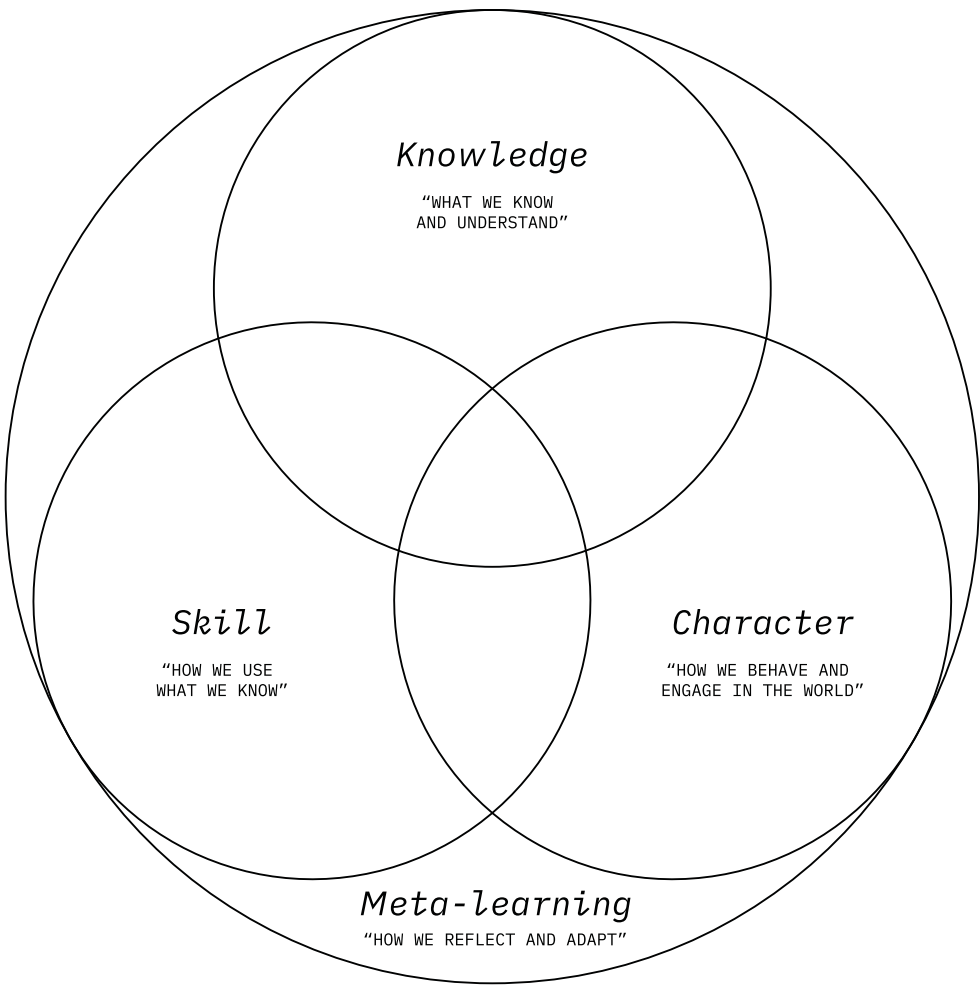


Figure 3. A holistic view of human capabilities. (Fadel & Groff, 2019)

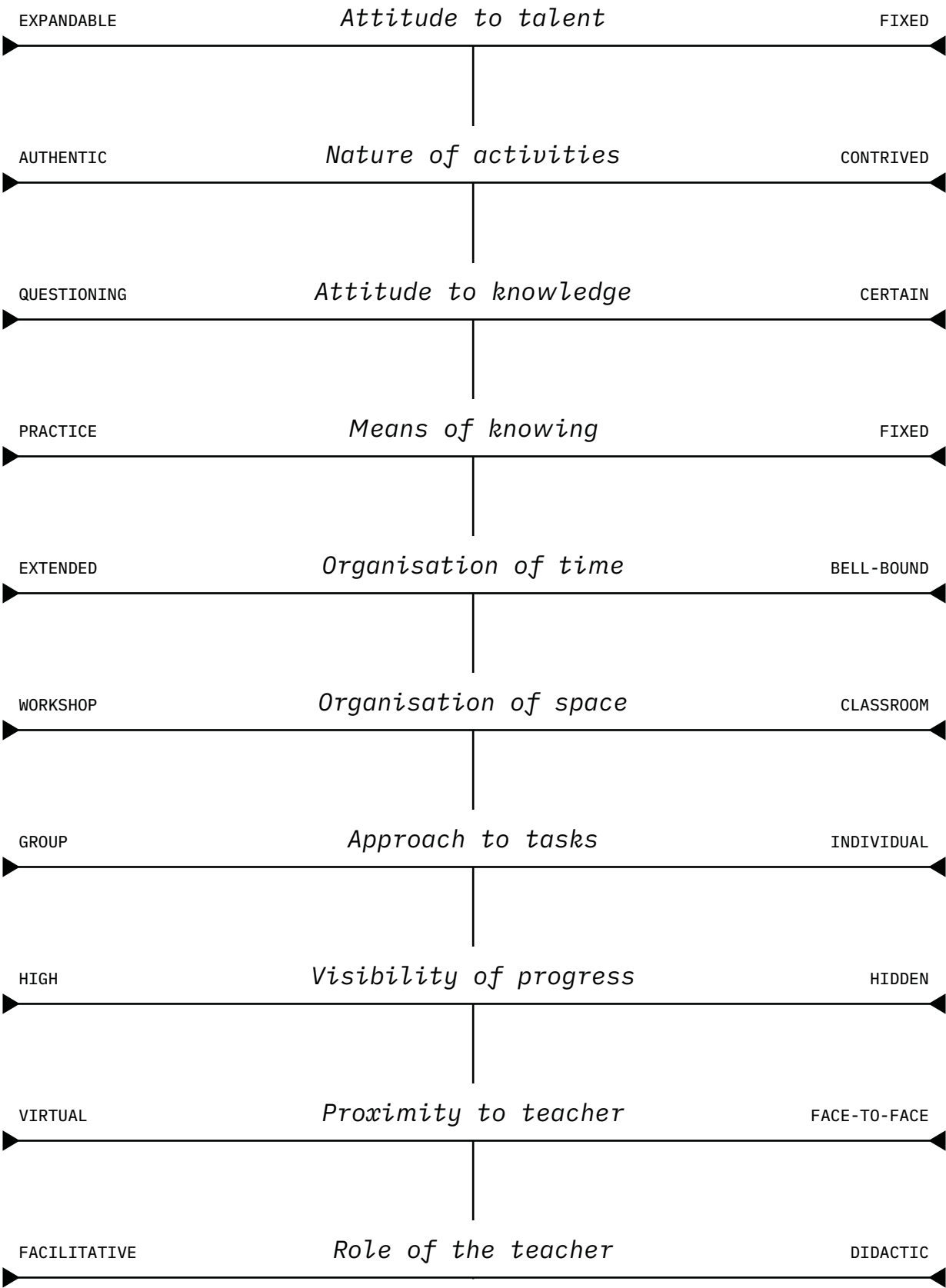


Figure 4. Signature pedagogies as a set of axes. (Lucas & Spencer, 2018)

2.2.3 EXPERIENTIAL LEARNING

Dieleman and Huisingsh define experiential learning as interacting with the world and seeing how it reacts. They summarise Kolb’s theory of experiential learning (Dieleman & Huisingsh, 2006; Kolb, 2015): experiential learning involves processing or “grasping” phenomena and intentional transformation of those phenomena to some desired end. Grasping and transformation can further be subdivided. Grasping can occur in the form of understanding or experiencing; transformation of those “grasps” can happen internally or externally. (Dieleman & Huisingsh, 2006) This is illustrated in Figure 5.

Understanding is the mode of grasping usually given the spotlight in educational systems. It is the analytical and rational grasping of knowledge. Experiencing, by contrast, is a mode of understanding that “relies on tangible and felt qualities of the immediate experience”. (Dieleman & Huisingsh, 2006) It grasps knowledge through lateral processes such as emotions, intuition and sensory data. Transformation through internalisation is the processing of grasped knowledge by reflection in a way that transforms our self-image and our image of the world. Transformation through externalisation is “active manipulation of the external world”. It is action directed out in the world, whereas internalisation is directed at our sense of self. (Dieleman & Huisingsh, 2006)

Kolb describes experiential learning as a cyclical process that feeds onto itself. Each type of learning mode gets activated during the learning process. They also point out that there are many similarities

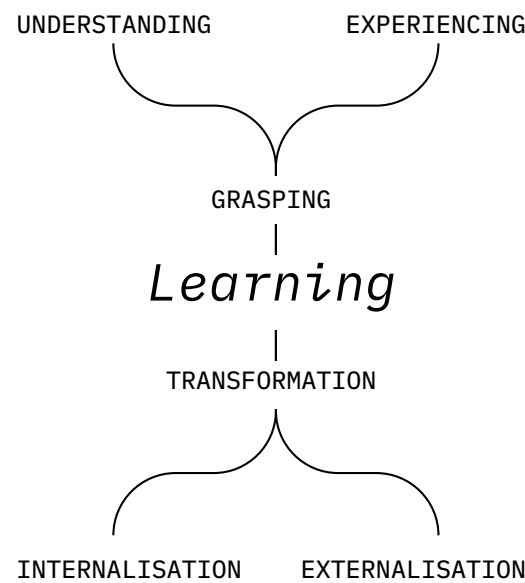


Figure 5. Aspects of experiential learning. (Kolb, 2015)

between learning processes and other “basic adaptive processes”, such as research, creativity, decision-making and problem-solving. (Kolb, 2015)

Dieleman and Huisingh summarise the different phases of experiential learning, illustrated in Figure 6. Firstly, the learner will have concrete experiences, either interacting with the world or through

models and theories, depending on the level of abstraction of the knowledge in question. Secondly, the learner will reflect on those observations and their relationship with that knowledge, transforming themselves. Thirdly, the learner will grasp and fit the learning into their existing knowledge frames. And fourthly, the learner will do something with that knowledge in the external world in the form of experimentation and action. This again results in experiences with the world.

2.2.4 TRAINING FOR ACTION

Crookall and Thorngate (2009) state that the connection between knowledge and action in Western education traditions has been broken, and action is sometimes even discouraged. In contrast, knowledge is put on a pedestal, exemplified by the emphasis on “knowledge transfer media” in academia, from lectures and tests to reading and essays. This same pattern repeats in scientific research: scientific consensus and knowledge of possible solutions are known long before anyone tries to implement them. (Crookall & Thorngate, 2009) This leads to inaction due to tradition and lack of practice.

Concerning Kolb’s theory of experiential learning, Dieleman and Huisingh state that learning about complex issues such as sustainability should especially focus on grasping by experiencing and transforming via externalisation as these two learning modes lead to enacting changes in the world. They further suggest that in the context of systemic learning, games would be the ideal vehicle for this. (Dieleman & Huisingh, 2006) Also, as was mentioned earlier, Wu and Lee suggest that first-hand experience dealing with issues sparks emotional triggers necessary for behavioural change. This can happen through games, as games encourage immersion in the experiences they provide. (Wu & Lee, 2015)

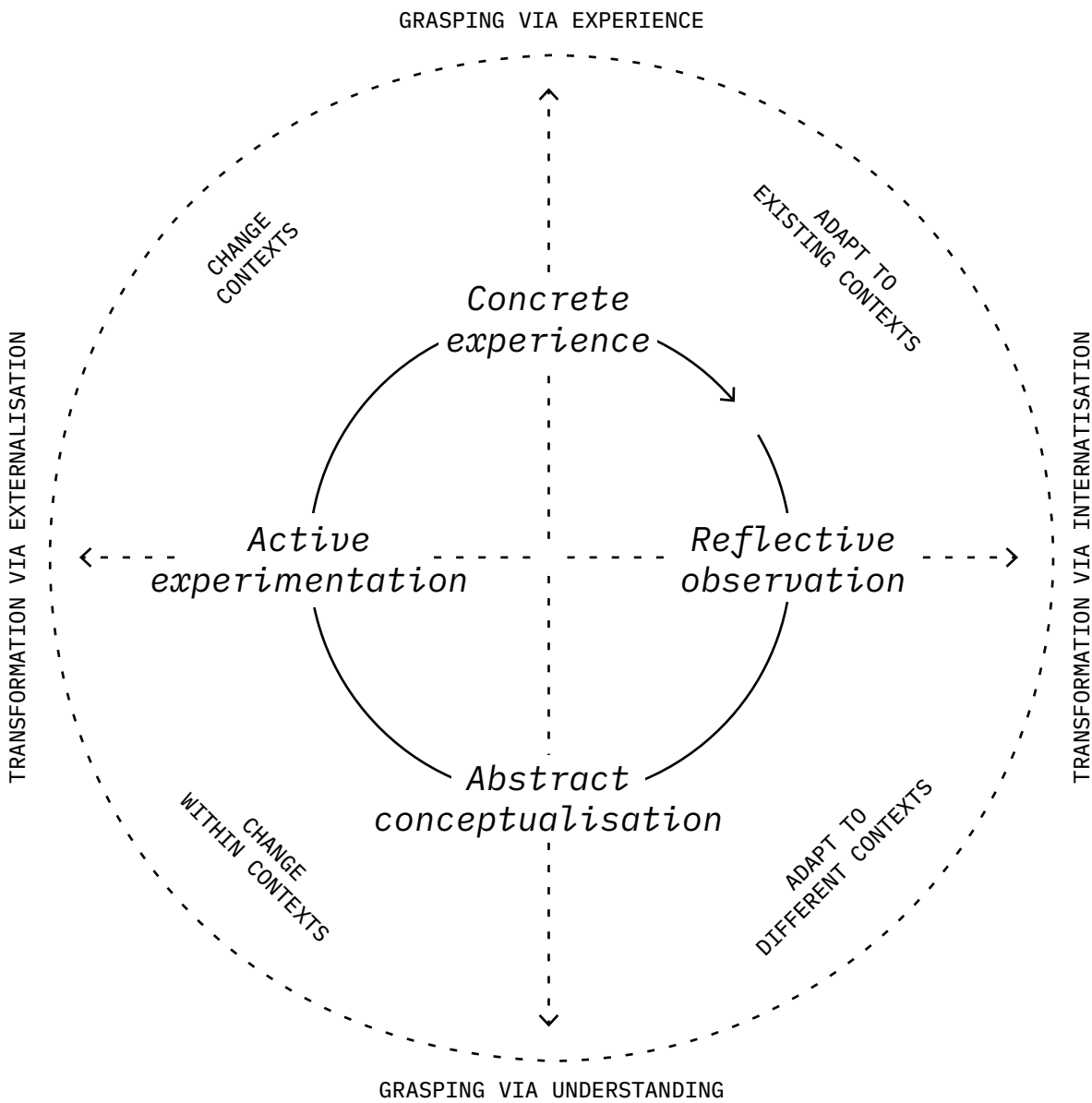


Figure 6. Phases of experiential learning. (Kolb, 2015)

2.3 SYSTEMS LEARNING GAMES

2.3.1 LEARNING THROUGH PLAY

Johan Huizinga argues in his widely known work “Homo Ludens: a study of the play element in culture” that for humans and other animals, play has been one of the primary methods for learning for millennia. (Huizinga, 2014) Huizinga uses an example of dog puppies brawling. They are not fighting for life or death; they know there are some limitations to their actions. Schell (2015) defines play as “manipulation that satisfies curiosity.”(Schell, 2015)* Play is a form of inquisitive action that allows players to learn through experience.

Play is also, by its nature, meaningful. This is what Huizinga characterises as the magic circle of play. Players inside the magic circle take anything that happens in a game very seriously, while to an outside observer, the play might have no meaning at all. Everyone within the magic circle understands that a game is being played, but players are still “playing along”. He argues that play happens because of an aesthetic need in humans to bring order into the world. People want to play to experience order and clarity, which the real world may lack. (Huizinga, 2014) This is illustrated in Figure 7.

Mackay (2013) argues that as kids go to school, play stops and so stops learning, suggesting that play is the primary method for learning for children. (Mackay, 2013) Interestingly, a lot of traditional learning can be classified as work, which is an antithesis to play according to Upton’s (2015) definition of play: “play is free movement within a system of constraints.” (Upton, 2015, 19) Work is not free nor constrained in a systematic fashion. Many educational games have tried to disguise work as play, with poor results. Klopfer et al. (2009) state that learning games should not be about tricking players into

* OTHER DEFINITIONS FOR PLAY ALSO EXIST. HUIZINGA DEFINES PLAY AS “A VOLUNTARY ACTIVITY OR OCCUPATION EXECUTED WITHIN CERTAIN FIXED LIMITS OF TIME AND PLACE, ACCORDING TO RULES FREELY ACCEPTED BUT ABSOLUTELY BINDING, HAVING ITS AIM IN ITSELF AND ACCOMPANIED BY A FEELING OF TENSION, JOY, AND THE CONSCIOUSNESS THAT IS DIFFERENT FROM ORDINARY LIFE.” (HUIZINGA, 2014) SUITS (2005) DEFINES PLAY AS “A VOLUNTARY ATTEMPT TO OVERCOME UNNECESSARY OBSTACLES.” (SUITS, 2005)

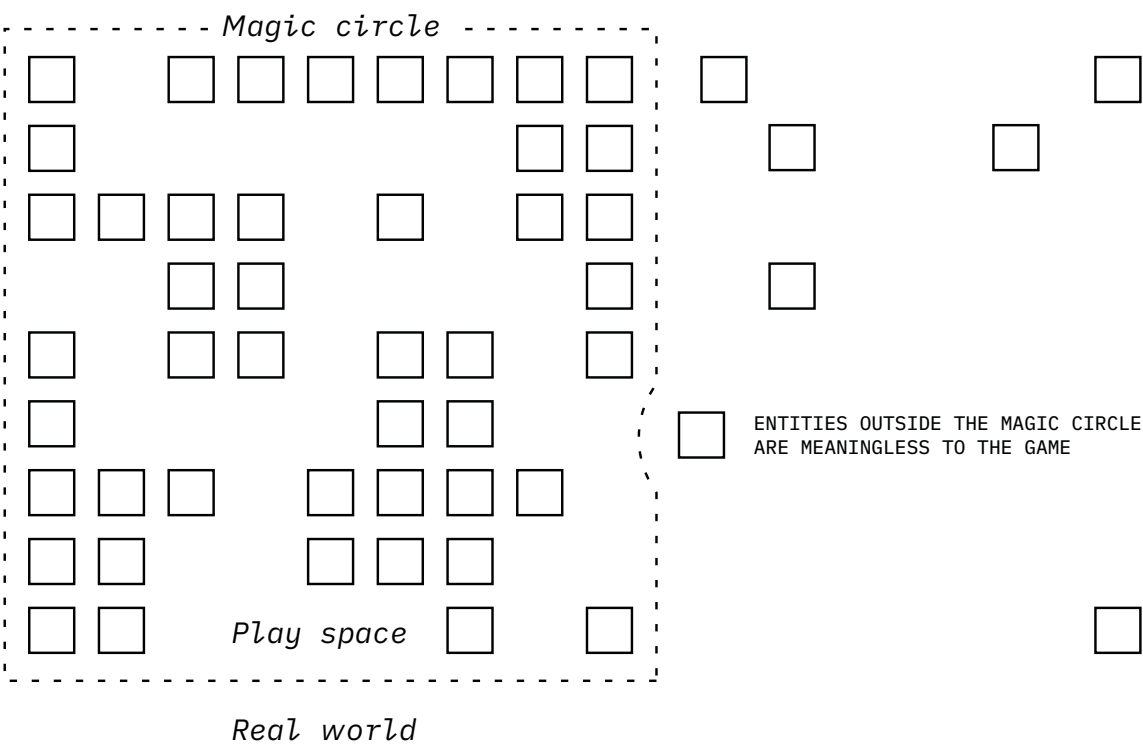


Figure 7. The Magic Circle (Huizinga, 2014)

learning by distracting them with entertainment, a factor outside of play, while also employing the same work-like learning methods applied in traditional school settings. (Klopfer et al., 2009, 32) Instead, players should be allowed to explore what’s inherently interesting and engaging about a topic through play. This will motivate players to go deeper into areas they are interested in, creating meaningful learning.

Therefore, any learning enforced from outside the magic circle will be disregarded. Instead, effective learning with games is inherent to the play and free of coercion.

Games are bounded play spaces. Play occurs within these bounds. The bounds can exist entirely within the common understanding of the players, such as with kid’s play, or be hardcoded into a digital game. Players participate in games by behaving according to these bounds, including formal rules and more loose principles of game etiquette. Players also push against them, testing the limits of the game and experimenting with what is permitted and what is not. This testing is transformed into “meta-reflective ‘questioning about’”. (Klopfer et al., 2009, 6) This corresponds to Kolb’s theory of experiential learning.

Similarly to system models, games can be modelled to represent imagined or real-world systems. There is little distinction between a simulation and a game: the one significant difference is that games implement play, thus allowing for experiential learning. Any learning must then be baked into the very fabric of the game.

2.3.2 LEARNING THROUGH CREATIVE FAILURES

Wright states that players in a game exist in a persistent network of interaction loops with the game and other players. Any action the player takes will affect the game state and other players, causing changes in the interaction loops and letting the player know how their actions affected the world. Wright continues that player progression happens through nested interaction loops of increasing complexity. This is illustrated in Figure 8. This drives players to get better. Players must be able to assert control or at least understand the dynamics of each interaction loop before progressing to a more complex set of interactions. (Lessons in Game Design, Lecture by Will Wright, 2003). If a player cannot reach the next level of interactions, they have failed. But failure is also feedback. Players will inevitably get better if there is something to learn from that failure. I would argue that the only effective way to learn through games is to fail multiple times in creative ways. Winning doesn't create any process of reflection on how things should be done differently to beat the challenge.

Challenging play drives players to acquire knowledge, skills, character traits or meta-learning skills that might help them perform better. Game designers can support this by making needed skills visible, but the motivation to acquire them already exists. Players struggle towards goals not because they want to build up their capabilities but to reach an intrinsically meaningful goal inherent to play. The secondary effect of the struggle is building capabilities to achieve the goal. (Upton, 2015) This is consistent with Huizinga's assessment of why play creates meaning.

Repetition in most games is inherent. Once you play a game through, you can start it all over again. This repetition enhances learning, especially if the game dynamics allow multiple routes to the end. The player can explore the "what if"s that rise during previous play-throughs, and by prodding the game systems from various angles, they accrue learnings. Crookall and Thorngate state that replayability can be consciously

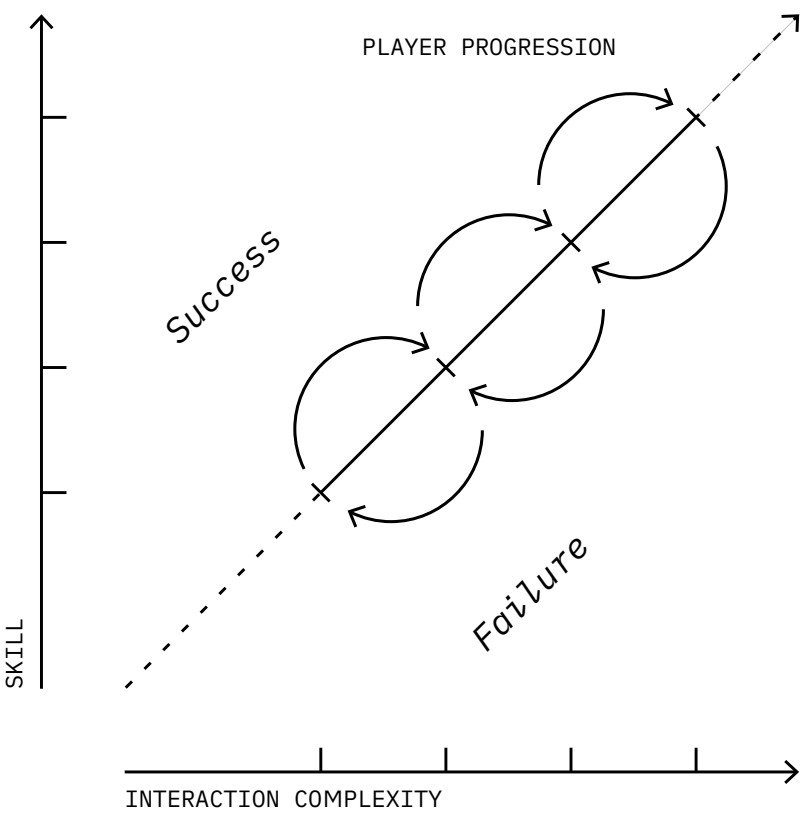


Figure 8. Nested success and failure loops. (Lessons in Game Design, Lecture by Will Wright, 2003)

leveraged for learning. Playing two games back to back, players get to learn from their mistakes and try again. (Crookall & Thorngate, 2009) To further leverage this, a retrospective can be arranged in-between games to quickly go through the learnings from the previous games for everyone to absorb during the second go-around. Despeisse (2018) further states that players often need multiple rounds to get familiar with the game-play, and learning can start. (Despeisse, 2018)

When actions in games are repeated within a highly responsive feedback loop in the context of challenging real-world problems, they can lead to lasting behavioural change outside games. Wu and Lee note that ample evidence suggests games can lead to behavioural changes. (Wu & Lee, 2015) A study by Ro et al. (2017) discovered that games show long-term impacts on behaviour, in this case, directed at sustainability issues, such as lowering energy consumption. They particularly highlighted the meaning of action as a tool to build awareness. (Ro et al., 2017)

2.3.3 TOPOLOGY OF LEARNING GAMES

As noted earlier, play is a natural tool for learning. Similarly, games have been used for learning for millennia. (Tasnim, 2012) According to Noemí and Máximo (2014), games and simulations were already developed for fighter pilot training during the 1940s. (Noemí & Máximo, 2014)

Educational games, serious games, game-based learning and gamification have been attempts to utilise learning in games. Unfortunately, many of these games have been unsuccessful. Based on my testing and research of various educational games, the main issue seems to be the lack of connection between learning content and play. This leads to poor games, with nothing to offer for learning. Players do not find them meaningful and disengage.

An example of this is the way some educational game designers use entertainment in the hopes of making the game engaging and thus making the learning content meaningful. The outcome in some cases, according to Hodent (2017), is to “put some cute animations on a quiz and instruct children more than truly educate them.” (Hodent, 2017) This is also known as “putting lipstick on a pig” or “chocolate-covered broccoli”. This is further exemplified by some developers describing the value of games in learning only as “adding entertainment” to an otherwise serious topic. (Katsaliaki & Mustafee, 2012; Folgado, 2014) In other words, to make boring content more enticing. Petché (2011) also states that educational games can turn a “potentially boring subject into something exciting and desirable,” (Petsche, 2011) which misses the point of using games for learning in the first place.

Commercial games are educational but not necessarily intentionally. Essentially every game is a learning game. (Järvilehto, personal communication, 02.09.2022) Games such as Europa Universalis 4, Civilization 6, The Sims or SimCity are highly educational purely because of their complex systems-driven approach to game design and a rich tapestry of real-life inspired content. Also, board games such as Eclipse or Terraforming Mars can be educational experiences in managing complexity. Still, I think none of these games would describe themselves as educational. Klopfer points out that many teachers utilising games opt to teach with these “commercial off the shelf” (COTS) games – as they are referred to in the educational games research community – because they are already well-made, balanced

experiences. The teacher only needs to consider how learning goals can be achieved within the game worlds. (Klopfer et al., 2009)

In recent years, there have been innumerable serious and educational games as part of research on sustainability or educational games, attempting to tackle the issue of teaching sustainability and climate change through games. Even multiple whole volumes on these topics have been published. The studies primarily focus on assessing implemented games or assessment of learning within the games. (Katsaliaki & Mustafee, 2012; Kaneda et al., 2021; Stanitsas et al., 2019) These could be classified as systems learning games.*

Beer Distribution Game (Goodwin & Franklin, 1994) and World Peace Game (Hunter, 2013) are two successful games in this category of games. The former was designed to teach managers about system dynamics by playing a game of managing a beer production system. They remarked:

The experiential nature of the exercise has proved to be very effective in illustrating not only how the underlying structure of the system produces outcomes but also how the system can control outcomes regardless of who is “managing” it. Invariably, the simulation dramatically demonstrates how powerful systems are and how important it is for managers to understand them. The ultimate objective, of course, is to show managers how to intervene in situations where systems thinking is appropriate and to exert the maximum leverage at their disposal. (Goodwin & Franklin, 1994)

The latter, World Peace Game, is a systems learning game for children, taught in a classroom over a two-month period. In it, four nations compete for scarce resources. However, the game can only be won if a list of 50 common problems is solved and the general well-being of each country is above the starting level. It powerfully immerses children in the world of complex systems. Based on the anecdotes in a book by Hunter, the method has been very successful. (Hunter, 2013)

* HERE IS JUST A TINY FRACTION OF OTHER ATTEMPTS AT SYSTEMS LEARNING GAMES (BOTTINO ET AL., 2016; CASTRONOVA & KNOWLES, 2015; CHENG ET AL., 2019; DE LA TORRE ET AL., 2021; DIELEMAN & HUISINGH, 2006; FABRICATORE & LOPEZ, 2012; FOLGADO, 2014; GATTI JUNIOR ET AL., 2020; GEE, 2009; KANEDA ET AL., 2021; KATSALIAKI & MUSTAFEE, 2012; NORDBY ET AL., 2016; PONTE ET AL., 2015; SHIRATORI ET AL., 2005; STANITSAS ET AL., 2019; TSAI ET AL., 2021; VARJUS, 2019; WEIJS ET AL., 2016)

3 THE EXPERIMENTS

During May and June of 2022, I designed six prototypes and many smaller experiments within three different game concepts exploring systemic topics and ways to bridge learning and play.

In Chapter 4, I will discuss various topics related to these themes, using the prototypes as examples. Here is an overview of the relevant ones. All of them were developed as board games, each focusing on a different topic. All are playable with two to four players. Each takes from 60 to 90 minutes to play through. Each of the presented games had two major prototypes, denoted as 1.0 and 2.0, respectively. Their differences are, in some cases, significant, and I will point those out where relevant. The design process took place in week-long sprints, with a week-long analysis of the previous prototype in between.

“Waste Land” is a land-management and capitalism game focusing on the poor balance between the economy and ecology. Players play as energy companies competing for the best land and suitable facilities on those lands to produce energy. The catch is that while non-renewable fuels are more reliable and effective in the short term, they produce waste, ultimately decreasing the efficiency of all facilities in polluted areas. Anything and everything can be traded in true capitalist fashion. Waste Land had two proper iterations, between which the game changed from an area control game to one emphasising negotiation and lateral thinking.

“Coalitions” is an engine-building game where you focus on making your society more productive than others until certain end conditions are fulfilled. It was my attempt at making a more abstract and modular systems learning game. Coalitions saw two major iterations of the game because I wanted to push the abstract and modular nature of the game to its edge. While the game was playable, it didn’t meet my standards and eventually ended up in the bin.

“Crisis” is a voting game that focuses on the political decision-making process on a very abstract level. Players collectively vote on various topics drawn from a deck randomly, attempting to gauge their decisions’ effects on their six common resource pools. Simultaneously each player tries to vote according to the requirements of their constituency. If any of the resources run out, they have failed.

In this thesis, I will not analyse these prototypes, their performance, chronological creation processes or other aspects of individual decisions made during the process. Instead, I will reflect on commonalities I spotted during the game design process and their effect on the elements that make these games meaningful for the players. I will then supplement these reflections with examples from the prototypes. Throughout Chapter 4, I will mainly use Waste Land 2.0 and Crisis 2.0 as examples. I will further introduce various mechanics of the games when needed.

4 REFLECTIONS

As discussed in Chapter 2, in order to utilise games in teaching about complex systems, learnings need to be sown into the very fabric of the game. Then learning will come as a byproduct of playing the game rather than at an attempt of distraction. In this part I will present some thoughts on how to design such experiences consistently.

In the game development community, one often hears that game design should be guided by the “Follow the Fun” guideline. This heuristic instructs that whenever you or your playtesters are “having fun” with your game, the game should pivot to enrich those aspects of the game that make it “fun”. While this approach has its uses, it also has its share of problems. Not only is this very superficial and an abstract principle, but it also is very difficult to define. It also risks driving the game farther from the original vision. Fun also doesn’t equate to a meaningful experience.

Flow theory by Csikszentmihalyi (1990) could work as a guiding principle for a designing meaningful game experiences, but as Upton argues, flow and play should not be equated, as play can be fun and satisfying even outside the “optimum experience” defined by flow theory. (Csikszentmihalyi, 1990; Upton, 2015) Moments of dullness and excruciating challenge are allowed in games and the player can choose to push those boundaries if they want to.

This part is split in two. First, I reflect on how any argument, be it a learning goal or a view of the world, should be communicated so that it passes through to the player in a way they will find meaningful. Secondly, I will reflect on how play spaces create possibilities for meaningful play.

4.1 ARGUMENTS

4.1.1 DEFINING “ARGUMENTS”

Ian Bogost (2007) argues that games are a tool for procedural rhetoric. Similarly to textual rhetoric, the game presents arguments about the issues represented in the game. Instead of using words, they are presented through interaction with the game’s systems. (Bogost, 2007)* Each game makes these arguments about the world, purposefully or not. The most frivolous game in the world argues that “being frivolous” is something worth spending time on. Dorn (1989) also notices that games are not value-free. (Dorn, 1989) They have an opinion about things that are presented. Arguments in games can also be thought of in context of learning capabilities: to be successful in a game, you need to know a certain thing, know how to do a something or play with certain type of mindset. In the game world these things are deemed valuable.

To induce learning in a player through a game, its arguments should reflect the wanted properties as seen through the world of the game. This could be as simple as: “In order to win, you need to be caring for others”. It is up to the designer to interpret this phrase through the lens of the game and its mechanics so that there is room for playfulness and meaning-creation by the player.

Although, because play inherently creates meaning, and play is inherently free of coercion, it gives the power to create meaning to the player rather than to the designer, even though game designers often have their views on how the game should be perceived and experienced. Players will map their experiences with the game to their own lived experiences and create unique meaning from those connections. This also happens with other cultural artefacts, but games are unique in that they allow for interactions and further manipulation of the world based on those perceived meanings, further increasing the potential for personal meaning for the player. Thus we as designers can only create the potential for arguments to be internalised by the player.

* ALTHOUGH BOGOST STATES THAT PROCEDURAL RHETORIC APPLIES ONLY TO VIDEO GAMES, I WILL BE USING IT IN THE CONTEXT OF BOARD GAMES, AS I THINK IT IS A VALUABLE TOOL FOR ANALYSING THEM.

4.1.2 SELECTING ARGUMENTS

Arguments and their style can vary depending on the purpose of the game. In a purely entertaining game, they can describe the types of experiences the designer wants the player to have. They can, for example, state that “This world is not a safe place”. But in games with ambitions beyond entertainment, such as learning games, the arguments can be the learning objectives the designer wants the players to achieve. For example, “Players learn the value of task management and coordination in a complex environment”.

For my prototypes, this process of finding appropriate learning goals was a traditional research iteration loop of finding what types of skills are considered essential to thrive in the 21st century and then going over the relevant literature on games and play and comparing which would be the easiest or most effective to induce with games. This formed a list of “arguments” that guided further game design work.

I tried to find skills that would easily transfer to other domains. This meant that character traits, such as adaptability and persistence, and organisational and lateral thinking skills, such as pattern recognition, planning and negotiation, got to the top of the list. In the end, this is the list of learning goals I tried to embed into my prototypes:

- Real-world Systems “The point is to stop a world-ending ecological catastrophe while simultaneously trying to stay on top of the game in a capitalist multipolar world, where other players’ motivations and actions are unknown and thus can’t be fully trusted, while also balancing and responding to internal pressures.”
- Systems Thinking “Game models reality on a scale that the player is not concentrated on the mundane operational actions, but can concentrate on how the system evolves and how to create good interventions into the system to make it work in your benefit, and ultimately for the benefit of all.”
- Transferable capabilities “Game leans on mechanics that encourage development of capabilities that are easily transferable to other

disciplines, such as compromise and negotiation, long-term planning and adaptability, pattern recognition and creative thinking, as well as coping with pressure and uncertainty.”

I further subdivided these as follows:

1. Real-world Systems - Survive and thrive in a multipolar world with what you’ve been given - Cope with not knowing other players’ motivations and actions - Stop global ecological catastrophe
2. Systems Thinking - Recognise patterns in system behaviour - Observe how the system behaves under various conditions - Create systems interventions
3. Transferable capabilities - Learn to make morally and economically sound decisions while under pressure and uncertainty - Learn effective collaboration and negotiation - Learn to observe patterns, plan just enough, execute swiftly and adapt creatively

4.1.3 MANAGING ARGUMENTS

I utilised the “Mechanics-Dynamics-Aesthetics” (MDA) framework by Hunicke (2004) to keep track of what arguments to present and how. MDA helps tie together the intended experience for the player and the mechanics that induce those experiences. (Hunicke et al., 2004) While it can’t dictate the player’s experience with the game, it helped me to manage and track wanted experiences and the experiments trying to induce them.

While the “official” MDA framework includes the stages of aesthetics, dynamics and mechanics to analyse your game, I wanted to add an additional step for arguments, illustrated in Figure 9. Arguments determine what experiences or “aesthetics” are expected. They can also determine what kinds of mechanics are used to represent them. This way, the arguments are a basis for everything.

In addition to mapping arguments to aesthetics and mechanics, I also mapped them to levels of decision-making and temporal game stages.

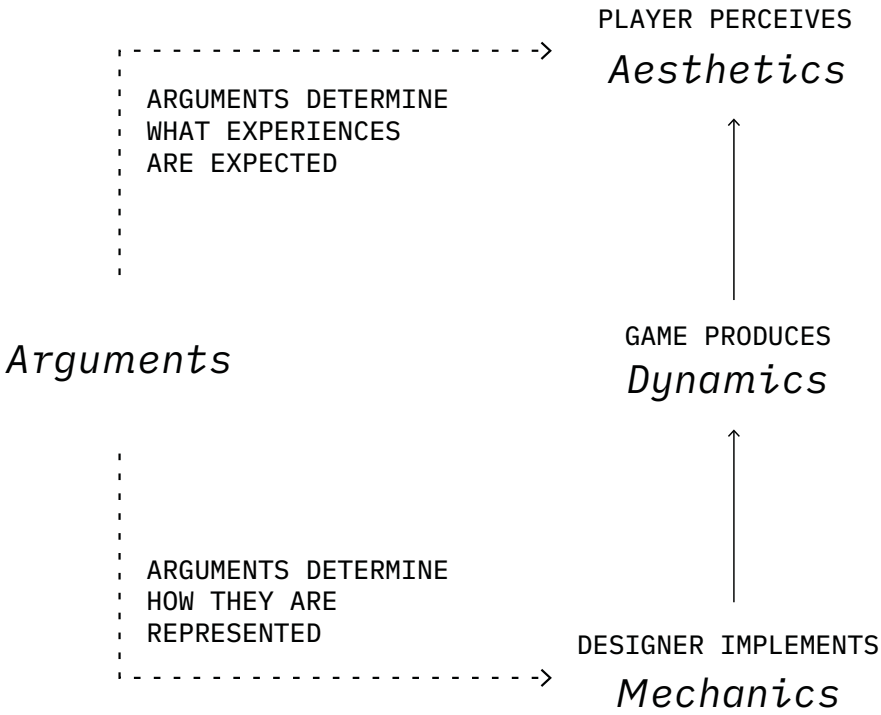


Figure 9. Arguments in addition to MDA framework.

Decision-making can occur on operational, tactical or strategic level. As a guideline for all my prototypes, I assigned the following concepts to each level:

- Operational: Uncertainty, decisions and competition
- Tactical: Collaboration for common good
- Strategic: Rigorous creativity

Temporal game states are here early, mid and end game. Depending on the arguments, the game may need to progress to some specific point before the dynamics representing the arguments become possible. In Crisis 2.0, players could, for example, play the early game manipulating others into voting for their benefit. But as the game progressed, these players would also drive their shared economy into imbalance, making it vulnerable to unforeseeable events. Thus in the mid to end game, the players often started to coordinate their actions

to avert catastrophes, shifting the game’s dynamics completely. The learning goals are further illustrated in Table 1.

4.1.4 IMPLEMENTING ARGUMENTS

An obvious way to introduce arguments into the game is by utilising them in the story or worldbuilding. In Crisis 2.0, worldbuilding was mainly constructed with the so-called “problem cards” that gave players three options to vote from regarding some problem. For example, one of the not-so-subtle cards read: “Insufficient safety protocols have led to a nuclear disaster”. The options to pick from were:

- 1. Offer disaster relief to those affected
- 2. Contain information about fallout spreading
- 3. Blame the accident on the neighbouring country

Each of these options modified some of the six different resources available. They presented a wide spread of worldviews to pick from, even though one was clearly better than the others. But depending on the situation in the game, you might have to cover up a nuclear disaster just so your economy won’t collapse. Through worldbuilding, I was able to introduce moral grey zones quite easily and exemplify the excruciatingly difficult decisions politicians might struggle with.

One powerful way to introduce arguments as mechanics is to bake them into the game’s goals. A goal embodies the meaning a system exists for. For example, in Crisis 1.0, a precursor to the voting game, players competed on who most effectively solved the most crises in their respective nations. Each problem was solved with a specified combination of “solution cards” that came in different colours. The solution cards were drafted in sets of three, where the topmost card of every set was visible to all players. Each crisis also had a turn limit within which the crisis needed to be solved, lest you get negative points. The earlier you could solve each crisis, the more points you earned. The end condition was met when all three shared global crises were solved. Because of the turn limit on each crisis, each

ARGUMENTS	LEARNING GOALS	AESTHETICS
REAL-WORLD SYSTEMS – UNCERTAINTY, DECISIONS AND COMPETITION		
THE POINT IS TO STOP A WORLD ENDING ECOLOGICAL CATASTROPHE, WHILE SIMULTANIOUSLY TRYING TO STAY ON TOP OF THE GAME IN A CAPITALIST MULTIPOLAR WORLD WHERE OTHER PLAYERS MOTIVATIONS AND ACTIONS ARE UNKNOWN AND THUS CAN’T BE FULLY TRUSTED, WHILE ALSO BALANCING AND RESPONDING TO INTERNAL PRESSURES.	SURVIVE AND THRIVE IN MULTIPOLAR WORLD WITH WHAT YOU’VE BEEN GIVEN	MAKE-DO, FIGHT OR FLIGHT, GO WITH THE FLOW, SHIFTING POWERS, PRODUCE, GATHER, CREATE, DESTROY
	COPE WITH NOT KNOWING OTHER PLAYERS MOTIVATIONS AND ACTIONS	UNCERTAINTY, HIDDEN, AMBIGUOUS, ON ONE’S TOES, BACKSTABBING, ESPIONAGE, POSSIBILITY OF DANGER
	LEARN TO MAKE MORALLY AND ECONOMICALLY SOUND DECISIONS, WHILE UNDER PRESSURE AND UNCERTAINTY	WHAT IS RIGHT?, INTERNAL VS EXTERNAL, NOW OR LATER, WHO IS A THREAT / ALLY?, WHO BENEFITS, WHAT ARE THE REPRECSSIONS?
SYSTEMS THINKING – COLLABORATION FOR COMMON GOOD		
GAME MODELS REALITY ON A SCALE THAT THE PLAYER IS NOT CONCENTRATED ON THE MUNDANE OPERATIONAL ACTIONS, BUT CAN CONCENTRATE ON HOW THE SYSTEM EVOLVES AND HOW TO CREATE GOOD INTERVENTIONS INTO THE SYSTEM TO MAKE IT WORK IN YOUR BENEFIT, AND ULTIMATELY FOR THE BENEFIT OF ALL	STOP GLOBAL ECOLOGICAL CATASTROPHE	MORAL DUTY, COMMON GOOD, COGNITIVE BIASES, LIMITS OF HUMAN PERCEPTION, FUTURE OF LIFE
	CREATE SYSTEMIC INTERVENTIONS (MEADOWS)	LIMIT, ESCALATE, CHANGE RULES, EVERYONE IS AFFECTED, WORK AT MULTIPLE SCALES, SHIFT THINKING
	LEARN EFFECTIVE COLLABORATION AND NEGOTIATION	AGREE, COMPROMISE, FORCE, COERCE, SECRET DEALS, INCENTIVISE, EMPATHY, WHAT OTHERS NEED
TRANSFERABLE SKILLS – RIGOROUS CREATIVITY		
GAME LEANS IN MECHANICS THAT ENCOURAGE COMPROMISE AND NEGOTIATION, LONG-TERM PLANNING AND ADAPTABILITY, PATTERN RECOGNITION AND CREATIVE THINKING, AS WELL AS COPING WITH PRESSURE AND UNCERTAINTY	RECOGNISE PATTERNS IN SYSTEM BEHAVIOUR	DYNAMICS, OTHER PLAYERS, BALANCE, WHAT CAN BE SEEN?, CORRELATION VS CAUSALITY
	OBSERVE HOW THE SYSTEM BEHAVES UNDER VARIOUS CONDITIONS	TEST, HYPOTHESIS, PROBE, EXPERIMENT, FAIL FAST
	LEARN TO OBSERVE PATTERNS, PLAN JUST ENOUGH, EXECUTE SWIFTLY AND ADAPT CREATIVELY	JUST-IN-TIME, PLANS FAIL, ISOLATE VARIABLE, 80/20, SOMETIMES INTUITION, FIRST HAVE SOMETHING TO OBSERVE

Table 1. Arguments, learning goals and aesthetics I was aiming for in my game prototypes.

player had to decide whether to focus on their own problems or if a global crisis was so close to the brink that they would need to devote actions from their turn to solve that particular crisis. This occasionally led to all players who were competing for the same “solution” resources, showing each other what they had on hand to coordinate and avert a world-ending disaster. This was meant to illustrate that even in a competition-driven world, actual change to global problems can only be achieved via cooperation, but it can often occur only at the last possible moment.

Emotions are an indispensable tool for crystallising arguments. Games are famous for creating powerful moments throughout the spectrum of emotions. This can be used as a tool to reinforce learning. Hunter, the designer of the World Peace Game, talks about how he had two warring student teams who suffered casualties on the battlefield write a letter to the homes of the fallen soldiers. This concretises the horrors of war, from the numbers on the screen to an individual, human level, and this experience will probably remain with the children for a long time. (Hunter, 2013)

In addition to worldbuilding and mechanics, arguments can be presented through the ambience or the game’s tone. Tone can take various forms, be it serious, light-hearted or satirical. While its creation may be more straightforward in other media, I’ve found that tone only emerges from some messy combination of interactions between the content and the dynamics of the game. Even though you can attempt to design for it, in my experience, tone only presents its true nature in playtesting.

This accidental discovery of tone happened with all of my prototypes. I didn’t necessarily have any plan in making the tone of the game anything specific at this point, other than I found the real-world systems that I drew inspiration from personally very broken. Almost without exception, the games felt like toeing the line of launching a cataclysmic chain of events. As described earlier, often the players would switch from hyper-competitive gameplay to collaboration to try to scramble back from the mess they themselves created. Even though these games weren’t satirical in my head, I suspect the shortened timespan with which events took place and their exaggerated effects on the game world made the game comical while also provoking serious discussions

about the systemic nature of the presented problems. This was a positive surprise and something I wanted to preserve in the game.

When dealing with topics as serious as this, humour can act as an icebreaker to get the discussion among players from general platitudes to analysing objectively what is wrong with the system. Satire and parody have been employed for centuries to depict broken systems by philosophical essayists and novelists such as Jonathan Swift (“A Modest Proposal”) or in modern times by the likes of John Oliver and South Park.

BYPASSING COGNITIVE BIASES WITH ARGUMENTS

People have a host of biases colouring their decision-making processes (Halpern, 2013). To see the real magnitude of phenomena, we need to be able to see through these biases. Games generally enable perspectives that are otherwise inaccessible to human reality. For example, games can compress time to better view the long-term effects of decisions, something that humans are notoriously poor at. (Halpern, 2013)

For example, In Crisis 2.0, time progressed in 10-year intervals. In the real world, voting on things as a representative of the people will not affect you personally because terms in office are relatively short compared to the time their decisions will effects the public. In the game, however, because the players stay in power indefinitely and see time progressing tens of years within one session of gameplay, they also get to see the direct consequences of your voting on the nation and its resources. Based on playtesting, the players could see the effects of their decisions clearly and it created some challenging moments and discussions in determining a balance between ethics, own gain and the good of the collective.

INTRODUCING CONSTANTLY SHIFTING LOYALTIES

Real-world systems often exhibit shifting loyalties. Partnerships and rivalries are forged based on the economic realities, quickly if needed. Designing such systems into games can have benefits in learning communications and negotiation capabilities.

In all my prototypes, I had two opposing goals for the players. The first was to stop the ecological or societal collapse, and the second was to gain success under the current economic and ideological incentives. I wanted to create an incentive structure where both competition and collaboration were not only viable strategies but also not mutually exclusive. In Waste Land 2.0, this was solved with shared land, whose pollution affected everyone on that land. Adding a robust trade mechanic also made everyone more friendly to each other as they wanted to get into the good graces of each other to get better deals. In Crisis 2.0, this was solved by adding global problem cards, individual agendas, and a shared resources pool. On top of these, each game acted as a common enemy for all the players. This forced players to shift their stance from time to time if the game itself was too threatening. In Waste Land, this was the waste mechanic; in Crisis, the shared resources pool.

4.1.5 PLAYERS SHOULD FIND MEANING THEMSELVES

Even though the arguments were well presented and implemented in a way that induces play, I think one crucial part of getting the message across is to leave the players to their own devices, to let them explore the game space at their own will and discerning meaning from the arguments that resonates with them.

For example, I left my moral judgement outside the game systems to allow the player to make traditionally “wrong” choices if they wanted to experiment with the game in that way. For example, the point of Waste Land 2.0 was to teach about various energy production methods, their environmental impact, and the profit incentives that make it challenging to transition to renewables. I didn’t want to, for example, make renewables gain more points for the player in exchange for lower income from the factories because that would impose a moral framework around the game and tell what the “right thing to do” is. Instead, I tried to model the actual benefit renewables would bring to ecology and the economy. Thus, while non-renewable factories would produce one waste per turn, renewables would produce none. Waste acted as a measure of pollution per area in the game. Once waste levels raised too high, the entire region would become toxic and stop producing until waste was removed to match sustainable levels. If all

areas became toxic, the game would end. This brought everyone together once in a while, even though the game is fundamentally a competition. In terms of economy, renewables would initially bring in less income per factory. However, the map had places where renewable energy would produce three times its original value. This was to mimic the location-dependent nature of renewables. Meanwhile, non-renewables could be built anywhere with no added bonuses.

This way, I could ensure that the point of the system comes across – non-renewables pollute but are highly profitable in the short term – without having to state it out loud. This also gives players room to experiment with both strategies or some mix of both.

4.2 PLAY SPACES

4.2.1 DEFINING “PLAY SPACES”

Upton’s implied definition for a play space is the entirety of possible states within the magic circle. (Upton, 2015) He uses the mathematical concept of phase space to illustrate this. Any parameter in the game is a dimension in the phase space. There is no precise way to visualise this, but for the purposes of discussion every change in parameters can be thought of as movement on a two dimensional surface. These movements can only happen within an enclosed space. Any movement is also constrained by previous states, and thus they have a “horizon of action”. Goals can be represented as a particular state within the sphase space. This is illustrated in Figure 10.

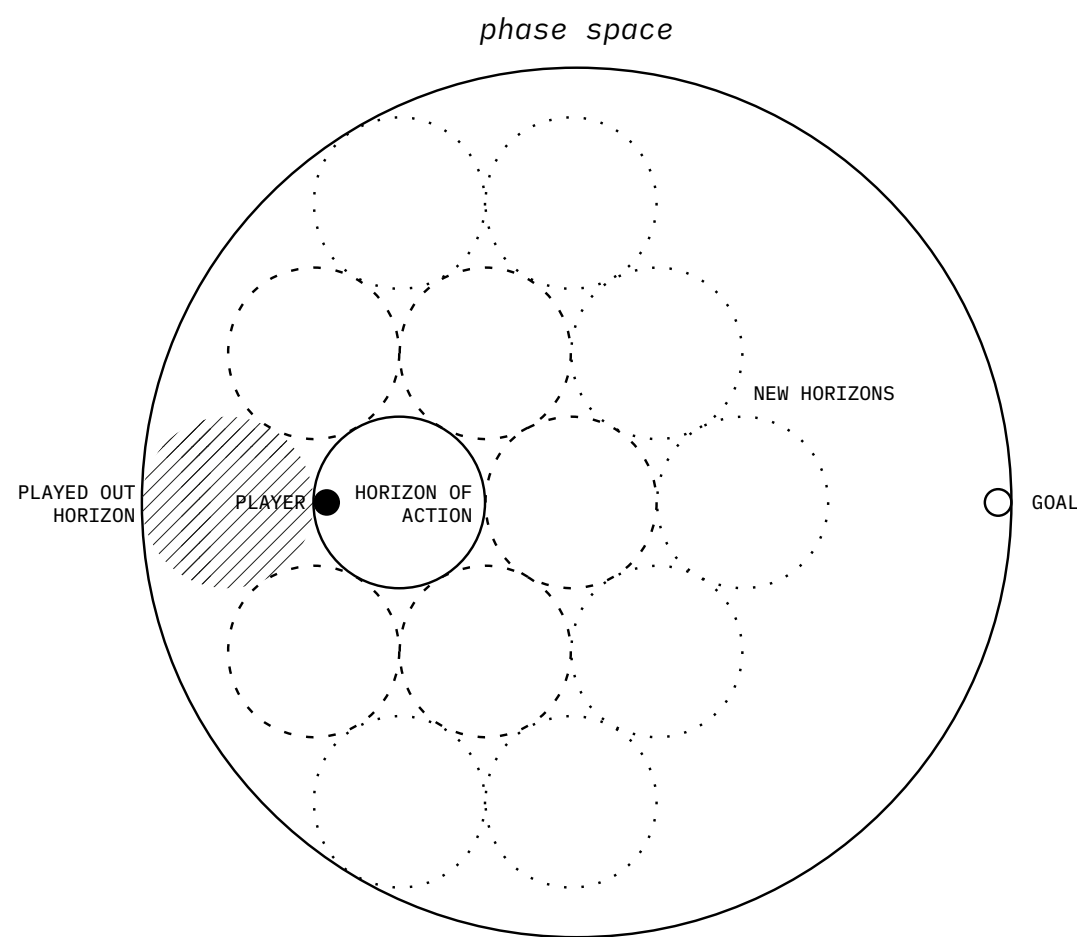


Figure 10. Phase space, goals and horizon of action. (Upton, 2015)

To create meaningful play spaces, Upton states that play needs to be constrained in a way that it falls in a sweetspot between boredom and confusion, frustration and satisfaction. In other words, play cannot be too easy or too hard and it cannot present too few or too many choices or points of data at once. Meaning arises from an optimal amount of complexity. Next I will detail two tools I used in the creation of my prototypes to constrain play spaces this way, while also maintaining resemblance to the underlying real-world systems these games were inspired by.

4.2.2 MEANINGFUL PLAY SPACES ARE COMPLEX

Wright echoes Upton in stating that complexity is born out of balance between chaos and order. Things are interconnected in systemic yet unpredictable ways. If the system does not have this balance, it can become a meaningless play space. If a system is too chaotic, its possibility space can diverge so much that no meaningful decisions can be made because the system is too unpredictable. Even the slightest variation to the input can cause a wildly different outcome. And on the other hand, if the system is too stable, there are no meaningful decisions to be made because the system is too deterministic and predictable. Whatever choice the player makes, the system will converge into the same position. (Lessons in Game Design, Lecture by Will Wright, 2003)

In Wasteland 1.0, I had this exact problem. The game was too deterministic. I knew from playtesting that once the player hits a certain threshold, they can’t lose the game unless they fail to notice the accumulation of waste and counteract it. This promoted only a single dominant strategy, which removed the play out of the game and made it “work”. The game essentially played itself.

But if the system’s complexity is calibrated just right, new balanced gameplay scenarios can emerge, that no one actually designed. A few good rules that allow for variability can breed complexity and depth without being hard-coded into the rules or mechanics. Adding additional rules or elements to increase complexity may increase the number of meaningful decisions, but they may also move the game away from, for example, social interactions and towards keeping track of rules and

work. Koster (2014) remarks that often, games that last the longest are those whose choice of only a few “ludemes”, or atomic gameplay components, offer the most depth of gameplay. (Koster, 2014) An excellent example of content emerging from complexity is Conway’s Game of Life, a cellular automaton programmed with only four rules, illustrated in the Figure 11. Depending on the starting configuration, these four rules can create innumerable things, such as self-replicating machines.

In Waste Land 2.0, I finally struck a good balance between chaos and order. In the game, players draw “lot cards” from a deck, determining where they can build during this round and “facility cards”, determining what they can build. Players will quickly fall into certain patterns determining what types of facilities they want and where they would prefer to build. However, there are many possible outcomes at any point in the game. Maybe luck doesn’t favour them, and they cannot trade for anything they need. Perhaps they find a golden opportunity to disturb their opponents, maybe throwing the area the opponent occupies over the waste threshold, rendering all of their facilities useless. Essentially, the horizon of action was wide enough to allow for multiple meaningful decisions in any situation, but not too wide to cause confusion.

LEARNING WITH COMPLEXITY

Complexity can be utilised for learning in various ways. For example, it can be set up to simulate poor prediction capabilities in complex

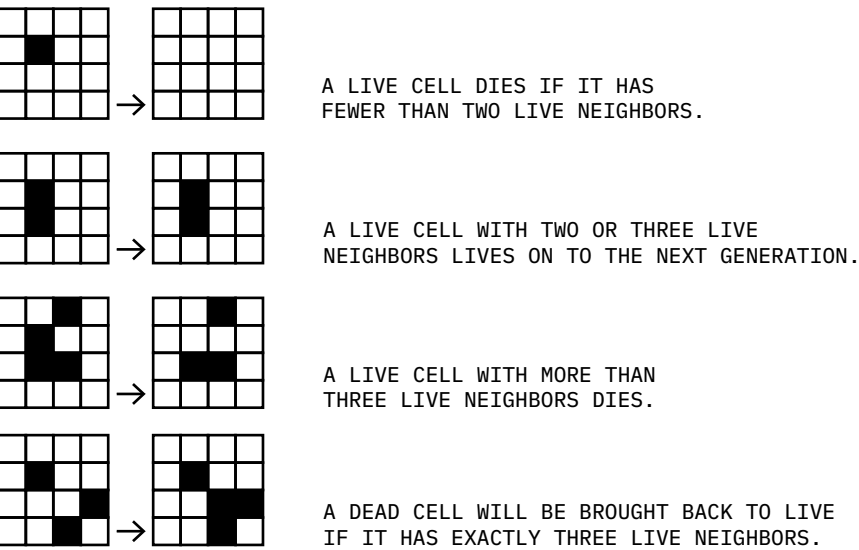


Figure 11. Rules for “Game of Life”. (Gardner, 1970)

real-world systems. This also encourages the players to play in an antifragile way, a concept by Taleb (2016): the player’s position only gets stronger from unforeseen, unlikely events that would otherwise devastate those building optimisation in their play. (Taleb, 2016) For example, in Waste Land 2.0, this can manifest in a way that players will distribute their production facilities all over rather than committing to a single mega-factory, that would gain other benefits. This is because if and when the waste threshold is exceeded, it would disable their entire production. However, they can avoid the severest form of this calamity with distributed production. Distribution also allows for a more diverse set of facilities to exist simultaneously, exemplifying one of the critical leverages outlined by Meadows, the self-organising property of complex systems. (Meadows, 2009) More diversity means on top of being more resilient to extreme events, they can also take many more opportunities if they are presented. These players don’t have to commit to a single factory type but can use whatever they have in their hand. This also gives them more flexibility when reacting to opponent manoeuvres.

Another way complexity stimulates learning is by its inherent variability. For the learnings in a game to transfer outside the game and into use in real-life situations, the game system needs to provide variable situations and complex cascading effects based on the player’s actions. In adaptive games, players have no other possibility for winning than to iteratively create and modify robust mental models based on their experiences with the game until the player can “dance” with the game systems, as Meadows put it. (Meadows, 2009) These mental models then transfer to real life. If the game is the same or similar at every playthrough, lacking the required complexity, the players cannot learn from their mistakes on a more conceptual level. Instead, they learn the particular manners the game operates and will eventually learn to exploit its weaknesses. This type of learning is tied to the game, rather than to transferable capabilities.

4.2.3 MEANINGFUL PLAY SPACES ARE DISTILLED

Often in games, faith in real-world systems they are modeling comes in the way of playfulness. Players may need to do repetitive tasks such as killing 30 boars or collecting ten pieces of wood because the game’s

systems have been modeled to mimic real-world systems to that resolution. Instead of providing new interesting decisions, you need to go out and do work where there are very few meaningful choices left.

For example, early in the design of Waste Land 1.0, I was still thinking of my game in terms of system modeling. This led me to think from the point of view of system realism: various mineral deposits would need renewal rates and maximum extraction rates. The terrain would need different modifiers dictating what can be built and where and with what efficiency. I had utterly forgotten the player experience. But as soon as I sat down to prototype the game, all of these parameters flew out of the window. One reason was that I had decided to stick with designing a board game, which would deny any possibility of complex simulation. But I also forgot to think if these parameters would help me build stronger arguments or if they promoted play, which they didn't. This drove me eventually to redesign the whole game into what became Waste Land 2.0.

Distilling games into the smallest possible set of mechanics or rules without shrinking its potential play space creates maximum opportunities for players to create meaning. Player meaning is created during free exploration within a system of constraints. (Upton, 2015) In systems learning games, instead of trying to replicate reality, which often results in adding work, game systems can be created at lower fidelity, allowing for more room for player imagination while simultaneously representing the system at sufficient accuracy. Thus in the case of real-world systems, they would not be duplicated but creatively translated into the most relevant version of the system, serving play and the arguments of the game.

LOW FIDELITY MINDSET

When it comes to the actual process of translating a system into a game, preferring low-fidelity representations provides a lot of benefits. In a nutshell, low-detail models promote a toy-like understanding of game systems for the players. (Lessons in Game Design, Lecture by Will Wright, 2003) It indicates this thing is meant for play. It doesn't mean that these systems are low in complexity. Instead, their level of detail is designed to promote a certain attitude towards the game and the malleability of its systems that enables systems learning.

Wright talks about getting players into the mindset of viewing the game as a model, setting the player's expectations accordingly. Suppose the game intentionally only uses representational and conceptual ways to model situations. In that case, the player will expect the game to behave like a model: only representing reality, not trying to mimic it. Human imagination can fill in the blanks left by the translation process. This lack of detail and resulting interpretability aids players to create the meaning of the experiences themselves. In a more symbolic model, the player can think about the model structure and underlying causes and effects rather than the real-world analogues and responses to a more specific model. (Lessons in Game Design, Lecture by Will Wright, 2003) Then again, a more "realistic" model, perhaps done with a higher fidelity thematisation, may promote learning in the context of that specific system. But in this case also, the system dynamics can remain low fidelity.

Wright also argues that the amount of detail in a model affects the player's ability to engage with it. If the level of detail is high, it is harder to discern the affecting properties of the model that create behaviour. It is also more challenging for the player to manipulate the model within the game without it causing unforeseen consequences. Low detail models also allow players with low skill levels to enter the game. If a user were tasked to design a house and provided with professional CAD software or even a pen and paper, most would be intimidated by the medium's lack of restrictions and possibilities. If they were provided legos to build from, the expected skill level and intimidation towards the task would be lower. Wright states that this method limits how much ignorance the player can display, making it easier to start playing around with. (Lessons in Game Design, Lecture by Will Wright, 2003)

The whole design process from Waste Land 1.0 to 2.0 felt like I was stripping away anything that didn't contribute to the learning goals. In 1.0, I had multiple different resource types for various purposes, which eventually collapsed into having one currency. In 1.0, I had agents who would physically have to move to areas, and the players would pick a resource cube from a sack to see if the land contained any resources. In 2.0, players were given random lots and possible facilities they could build. All these changes brought the players closer together, from an experience reminiscent of multiplayer solitaire to a collaborative and interactive experience, where each player would be concerned about everyone else's

actions. And when it was all done, the game was much more fluent and enjoyable. In Figure 12 I plot and compare my game prototypes to other systems learning games in their amount of detail and abstraction.

Finally, these factors are all malleable, at least in video games. They can be dynamically adjusted as the player progresses and gains confidence and skill in the tasks required. This could be compared to well-executed tutorial systems, where the complexity of the player’s toolset is steadily increased throughout the game instead of starting with every tool.

UTILISING FICTIONAL SETTINGS

One major problem with strict translations from a system to a game is that they do not allow any deviation from the actual mechanics that make the system work in the real world. If any deviation happens, the player may feel cognitive dissonance because they expect the system to behave as in the

real world. To get around this, one solution is to introduce fiction into the system structure on purpose. By clearly deviating from the expected format and presenting the system, for example, in a fantastical setting, any preconceived notions of how the game or system should work disappear. This won’t prevent the learnings from being gained from the game, as long as the fantasy doesn’t distract from the underlying arguments.

In Crisis 2.0, I framed the game theme around a hegemonic superpower controlling everything on a fictional Earth. This deviation from reality allowed me to remove a lot of ambiguity that wasn’t central to the argument of the game and instead focus on the core mechanics of policymaking concerning resource management and the ethics of various decisions. If I had illustrated the world as it is with individual nations, I would have had to deal with additional questions arising from that, for example, the adherence to policies created by this consortium by respective countries, such as within the UN or EU. To be clear, this only worked because players are most likely familiar with the basic concept of voting. This approach might not have worked well if the mechanics were derived from a more involved policymaking process.

4.2.4 SELECTING THE RIGHT SYSTEM TO TRANSLATE

Carefully choosing what system or part of a real-world system to represent in a systems learning game is essential as this one decision dictates the rest of the game mechanics development. I find that often the right mindset in working out what the right system would be is crucial. Instead of asking, “how do I represent this as a game?” asking, “is this representable as a game?” lets me discover and discard topics much more easily.

Furthermore, common knowledge is usually blind to the more nuanced dynamics of a system under the hood. Carefully studying expert literature on the subject can powerfully illustrate the main drivers behind any particular system and open avenues for creative system translation and condensation. Suppose expertise of systems knowledge is reflected in the game dynamics. This will bring credibility to the game and new learnable content to players already familiar with the system dynamics. My prototypes and their themes revolved around energy, policy and their intersection. I’m by no means an expert on any of that. To get inspired about the right things as well as to give an accurate representation of these systems, I observed

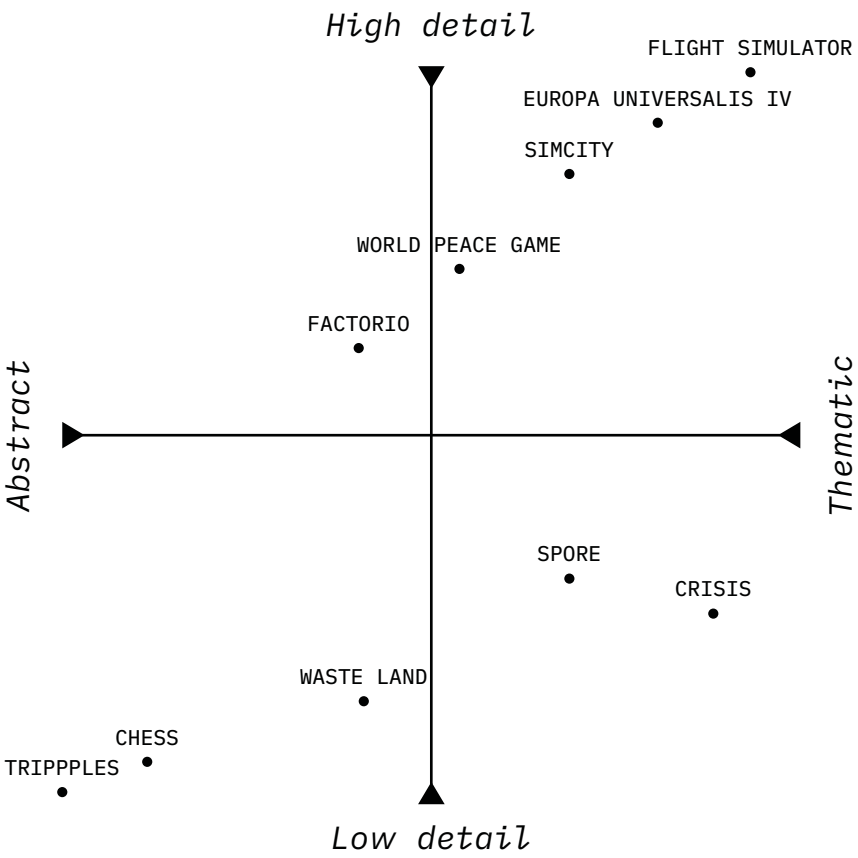


Figure 12. Systems learning games plotted manually, comparing level of detail and amount of abstraction.

courses on environmental policy and economics, skimmed through Vaclav Smil’s literature on energy and energy transitions, as well as read through the European Union’s climate goals & list of policies over the years.

One method to narrow the translation scope is to first represent the systems as a graph at various fidelities. For my project, I researched multiple systems models in which these systems were already illustrated and modified for my purposes based on the arguments and the level of fidelity I needed. The final flowchart I used as a reference can be found in Figure 13. This helped me focus my design decisions on the system’s most essential components. All else could be discarded.

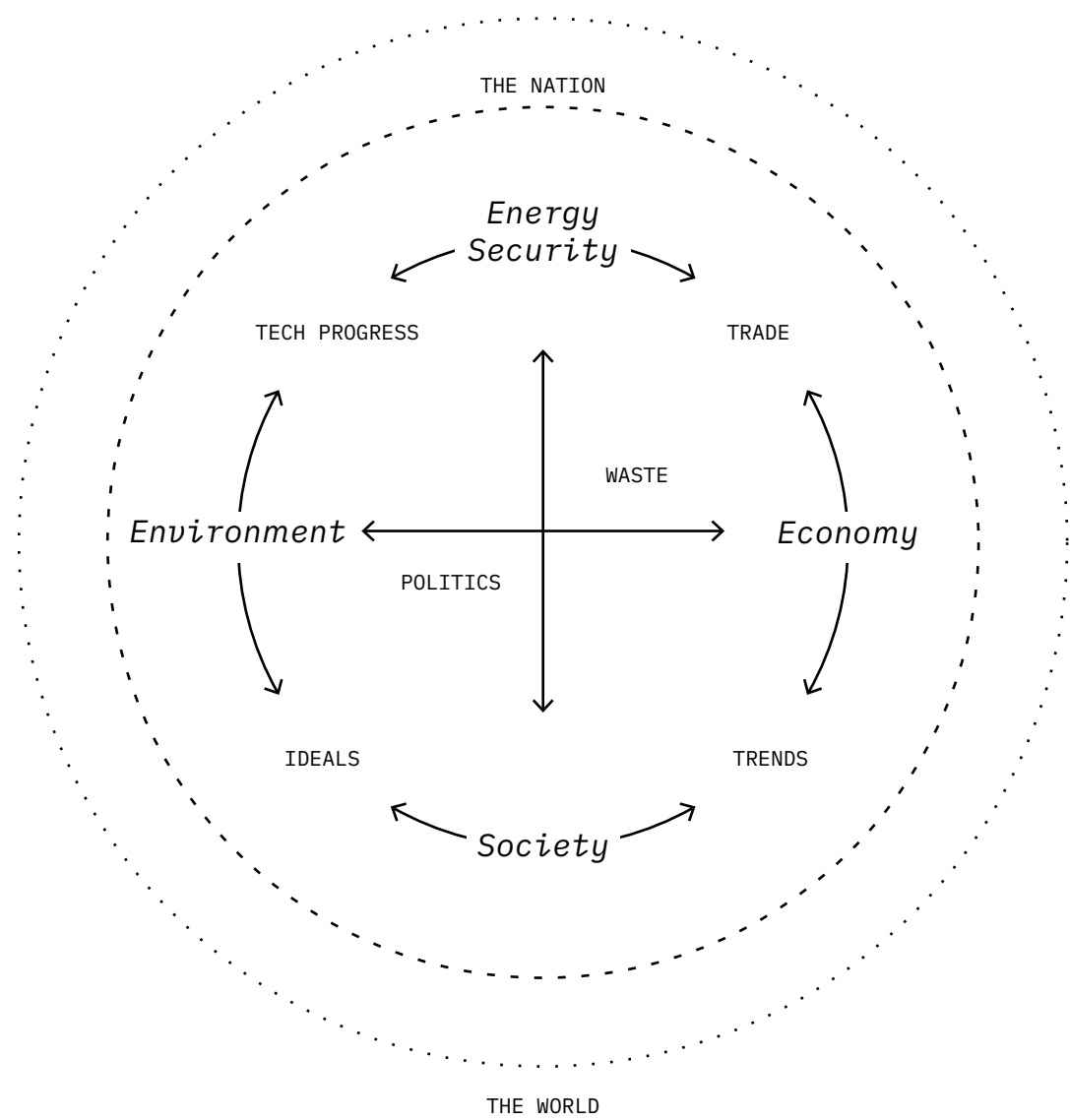


Figure 13. Flowchart depicting energy security, environment, society, economy and their interactions on a very abstracted level.

5 DISCUSSION

As stated in the introduction, my research questions were as follows:

- How to design arguments that players will find meaningful, be they learning goals or other arguments about the world?
- How to design play spaces that encourage players to build capabilities?
- How to design games that generate understanding and skill to thrive in complex systems?

I successfully discovered ways to create ways that at least theoretically bridge the concepts of arguments and player-created meaning through the employment of rhetoric and play spaces. I also outlined specific examples of cultivating capabilities necessary for acting within complex systems, such as awareness of one's cognitive biases and negotiation ability. These are by no means an exhaustive list of techniques but rather an exploration into how something like this could be achieved. Evaluating these findings is difficult as my prototypes have only been tested with a handful of people, whereas assessing learning can be arduous and lengthy. But in terms of evaluating whether these techniques created meaningful gameplay, I can say that the players involved enjoyed them and found themselves immersed. But again, further testing is needed to determine if this statement is generalisable across a larger cohort.

THOUGHTS ON APPLYING LEARNING GAMES

The idea of simply porting existing educational models or ways of working in the classroom to games must be thrown out, and games should be examined seriously in terms of what they can offer in the educational field that is unique to games. This is true of all media. The media should be chosen based on its merits and potential to distribute the wanted arguments in a meaningful way instead based on their popularity among some people.

It should also be acknowledged that games are not the ultimate silver bullet in education but a learning tool among others. Games are not as good as books in portraying textual information. Similarly, books are not great at showing information through interactivity. In my opinion, games cannot replace everything traditional learning methods can offer. Still, I think games can provide learning tools for the whole spectrum of human capabilities, most of which have long been ignored by the conventional schools of thought in pedagogy: human capabilities are more than just knowledge; learning skills, character and meta-learning also need nurturing. They are taught in schools today, but they seem to be subservient to the requirements of learning knowledge. But knowledge is only as valuable as the ability to utilise it in the right place at the right time. Knowledge is irrelevant if the student doesn't develop characteristics to be persistent and adaptable in their pursuits or doesn't have the right tools to learn in a way that suits them best.

Games can also be problematic in their ability to persuade and affect behaviour. They employ rhetoric, whether their authors admit it or not. Media has been used to manipulate opinions, and games also have this ability. Careful assessment of the arguments the game and its emergent systems are making is of the utmost importance. Complex systems are unpredictable; in some cases, it's impossible to map every possible permutation within them. Many games also encourage predatory behaviour: the gamer community around hyper-competitive games has long been a well-known example of widespread toxicity. Thus hyper-competitive attitudes should be balanced with games that promote cooperation and the well-being of a whole system.

FUTURE AREAS FOR EXPLORATION

To continue this research, I would need to build another prototype based on the learnings gathered while writing this thesis and test it on various cohorts. An important aspect highlighted by many writers (Dieleman & Huisinigh, 2006; Dorn, 1989; Eisenack & Reckien, 2013; Taha, 2017) in terms of ensuring learning with games is the utilisation of debriefings after the game experience, as well as some methods for assessing learning, such as the Stealth Assessment method. (Shute & Ventura, 2013) There are multiple directions this project could go. Video game implementation is also one possibility.

6 CONCLUSION

Counterintuitive, complex systems and their mismanagement causes systemic problems that are felt globally. The world needs people with tools to effectively solve these problems. System properties and ways to affect them are difficult to learn with traditional methods. Systems models are one way to learn about systems. Systems interventions however also require a more emotional and specific connection to the topic and these models are not accessible to normal people, nor do they produce emotional responses or train in proactive action. Learning through experience is a well-researched phenomena for training action. Games have inherent properties making them optimal for training action. They train properties in player's that are conducive to self-directed problem-solving empathetically.

Systems learning games communicate central arguments about systems and their issues by implementing them in an interactive format, letting players experience overcoming adversity within the systems, thus learning about system interventions through action. A plethora of systems learning games exist, many focusing on the topic of sustainability, but very few texts have been written about the gameplay within specifically these kinds of games and designing for gameplay that meaningfully trains their players in set learning goals.

I explored some methods of inducing meaningful play in the context of system learning games.

Firstly, learning goals can be seen as arguments presented through procedural rhetoric. To connect arguments to players and make them meaningful, they can be mapped using the MDA framework and implemented using worldbuilding, goals or as more pervasive elements in the game, such as the theme or tone. Players should be left to their own devices when discovering these meanings.

Secondly, in terms of represented systems and resulting play spaces, instead of trying to replicate reality, systems can be translated to induce play. Systems should be sufficiently complex to generate meaningful decision-making. They should focus on crafting situations where the player agency is at its peak.

Finally, carefully selecting what system to represent in a game is crucial, as every consequent game design decision depends on this question. Flowcharts and other low-effort tools should be employed to scope the playability of the system representation without committing to it.

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FIGURES

Figure 1. List of systems leverage points, impact amount of a leverage as a function of effort required (adapted from Meadows, 2009)

Figure 2. Incompatibility of models. (adapted from Lessons in Game Design, Lecture by Will Wright, 2003)

Figure 3. A holistic view of human capabilities. (adapted from Fadel & Groff, 2019)

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